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Energy Efficient Adaptive Bit-Mapping Based Collision-Free MAC Protocol for VANETs

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ABSTRACT

In the present work, an energy-efficient medium access control (MAC) protocol is proposed for the adaptive data traffic condition in VANET. The proposed adaptive collision free MAC-bit map assisted (ACFM-BMA) protocol represents a significant advancement in vehicular communication systems by integrating strategies focused on electric vehicle (EV) battery levels and buffer status. This protocol optimizes communication efficiency and energy conservation through a holistic approach that considers the dynamic nature of both the vehicles' energy resources and their data buffering capabilities. By incorporating real-time data about EV battery levels, the protocol ensures that communication tasks are scheduled in a manner that conserves energy, extending the operational lifespan of vehicles in the network. Additionally, the protocol takes into account the buffer status of each vehicle, preventing data congestion and loss and enhancing the reliability of data transmission. The integration of battery and buffer status information enables more efficient use of the communication channel, reducing idle times and improving overall throughput. The performance of the proposed method is analyzed based on overall vehicle density, the number of vehicles generating periodic data traffic, and the event occurrence probability of vehicles generating event data traffic. Three different scenarios are considered: low, medium, and high data traffic, corresponding to varying numbers of vehicles. The results demonstrate that the proposed method significantly conserves energy across all scenarios.

1 | Introduction

In contemporary times, a burgeoning demand exists within the automotive industry for electric vehicles (EVs), primarily attributed to their environmentally friendly attributes compared to traditional petrol and diesel vehicles. This surge in EV popularity has ushered in a realm of new opportunities, notably with the increased availability of onboard computation and communication capabilities, vehicles are moving towards automated driving and intelligent transportation systems. Intelligent transport systems (ITS) are integral to the evolution of urban transportation, holding a crucial position in the age of technology. ITS deployments can offer the following benefits: improved safety, efficiency, mobility, accessibility, and intermodal connections [1]. Adapting technologies such as IoT, edge intelligence, 5G [2], and blockchain in vehicle architecture has increased possibilities towards efficient and sustainable transportation systems [3].

The advancement of EV technology has facilitated vehicle-to-grid communication capabilities, enabling interactions with the electric grid for bidirectional energy flow, grid stabilization, and demand response purposes. Additionally, the incorporation of advanced driver assistance systems (ADAS) and autonomous driving functionalities underscores the heightened significance of EV communication protocols.

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Vehicular ad hoc networks (VANETs), with their fast-moving nodes and dynamic topology, capture researchers' attention amidst rapid urbanization. Scholars are increasingly drawn to the field, driven by the quest for safer and more comfortable driving experiences in the future. It consists of both inter-vehicular vehicle-to-vehicle (V2V) and intra-vehicular vehicle-to-infrastructure (V2I) communication [4].

The IEEE 802.11p [5], also known as dedicated short-range communication (DSRC), 802.11p is an upcoming WLAN standard intended for future traffic safety systems. This is currently the only standard with support for direct vehicle-to-vehicle (V2V) communication [6]. The 802.11p protocol operates on a carrier sense multiple access (CSMA) basis, where nodes listen to the channel before transmitting. 802.11p functions as a contention-free protocol, eliminating competition for the wireless channel during communication. However, this results in inefficient bandwidth utilization, especially in high-density traffic scenarios. To optimize network efficiency, medium access control (MAC) protocols must incorporate mechanisms that intelligently allocate resources while considering energy constraints. One such crucial metric is the state of charge (SOC), which determines the available energy capacity of a battery before reaching its end-of-discharge (EoD) time [7]. Integrating SOC estimation into the MAC protocol enables energy-aware scheduling, prioritizing transmissions from nodes with lower energy levels to enhance network longevity [8].

In discussions concerning MAC, contention-based protocols are generally favored over contention-free protocols [9–12]. Eliminating unused transmission slots in a contention-free MAC protocol is difficult [13], while in the contention method, it is organically dispersed and adjusts to traffic loads [14]. The protocol should prioritize the transmission of information so that it does not exceed the battery's operating safety level (i.e., the level at which the battery suffers irreversible damage to its chemistry) [8].

In critical conditions, energy efficiency becomes critical challenge for EVs in VANETs, not just for propulsion but also for ensuring reliable communication. For example, if an EV's battery is nearly depleted in a remote area, it must still communicate distress signals to RSUs or nearby vehicles. Similarly, in case, if an accident occurs in a high-traffic zone, an EV with low battery must be able to transmit emergency alerts to avoid further collisions. During the extreme weather conditions like storms or freezing temperatures, rapid battery drain can compromise communication, making energy efficiency essential. While EVs have dedicated battery reserves, these batteries power multiple subsystems, including vehicle sensors, communication modules, and other peripheral components [15]. Continuous data transmission without an optimized MAC protocol can lead to unnecessary energy consumption, reducing the availability of power for essential vehicle functions. More importantly, in situations where an EV's battery is nearly exhausted due to unforeseen conditions—such as prolonged travel, adverse weather, or charging delays—maintaining efficient communication becomes even more crucial. If an EV is unable to transmit emergency messages, participate in cooperative driving coordination, or send distress signals to RSUs, it can lead to serious safety risks, including traffic congestion, accidents, or vehicle breakdowns in unsafe locations.

An energy-efficient MAC protocol ensures that communication resources are utilized optimally, reducing transmission overhead, minimizing power drain on communication modules, and prioritizing critical message dissemination when battery levels are low. By preventing unnecessary re-transmissions and idle listening, an efficient MAC enhances network reliability, ensures seamless data exchange, and supports emergency scenarios, making it a necessity for sustainable and safe EV-VANET operations.

To address the all above concerns, in the present work, we have proposed an adaptive collision free MAC-bit map assisted (ACFM-BMA) protocol that prioritizes energy efficiency and integrates state-of-charge awareness to address the aforementioned issues. This protocol operates on a hybrid approach, incorporating the electric vehicle's SOC information and buffer status to enhance channel utilization and energy efficiency.

The paper is organized as follows: Section 2 reviews existing vehicular communication protocols, highlighting the limitations of IEEE 802.11p and the advantages of TDMA-based protocols while identifying challenges in current V2V MAC protocols. Section 3 details the methodology of the proposed ACFM-BMA protocol, explaining its time frame and slot structure, prioritization based on battery and buffer status, and providing equations for energy consumption. Section 4 presents the performance analysis of the protocol across different traffic scenarios, demonstrating significant energy savings. Section 5 concludes the paper by summarizing key findings and suggesting future research directions.

2 | Related Works

IEEE 802.11p, an approved amendment to the IEEE 802.11 standard, introduces wireless access in vehicular environments (WAVE), a communication system for vehicles. It provides enhancements to the existing 802.11 (WiFi) to support intelligent transportation systems (ITS) applications.

802.11p improves upon the shortcomings of its precursor, 802.11, and employs CSMA/CA as the primary medium access mechanism within the WAVE MAC. Nonetheless, CSMA faces difficulties in ensuring adequate quality of service (QoS) for inter-vehicular communication due to the dynamic topology and dense node population, as given in [16].

TDMA-based protocols serve as a better alternative to their CSMA counterparts. The TDMA MAC protocols for safety applications of VANETs are classified into six categories, including multi-channel coordination, clustering, TDMA, SDMA, self-organizing TDMA (STDMA), and cognitive radio [17]. The authors have thoroughly studied and analyzed different MAC protocols, evaluating their potential applicability in future contexts. Among these, the ACFM-TDMA protocol has been proposed as an effective TDMA-based approach for vehicular networks [18]. ACFM-TDMA dynamically adjusts slot assignments based on traffic density, where RSUs play a key role in scheduling and disseminating control information. By adapting the slot allocation cycle, ACFM-TDMA ensures improved channel utilization and fairness, making it suitable for vehicular communication scenarios with varying traffic conditions. The dynamic nature of ACFM-TDMA

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enhances packet delivery reliability and reduces delays, addressing key challenges in VANETs. However, it is identified that the authors have not addressed the energy consumption issue which needs further analysis.

The author of [19] provided an overview of numerous V2V MAC protocols, including various VANET standards. The authors focused on examining the advantages and limitations of these MAC protocols. Their findings emphasized several challenges that must be addressed in future research to facilitate the implementation of highly efficient MAC protocols for VANETs. These challenges include:

- Contention-free protocols can meet the QoS requirements for real-time applications, but they require significant coordination due to VANETs' unique characteristics.
- Contention-based protocols are not suitable for critical safety messages due to their unpredictable delay and poor performance in densely populated networks.

The proposed ACFM-BMA protocol merges strategies by considering EV battery levels and buffer status, aiming to optimize communication efficiency and energy conservation in-vehicle networks. This approach combines data about electric vehicle batteries and buffer space availability to make the most of the communication channel while also saving energy. By leveraging these insights, the protocol enhances overall performance, ensuring a balance between effective data transmission and resource utilization in vehicular communication systems.

3 | Methodology

This research builds upon the ACFM-TDMA protocol, which dynamically adjusts time slot assignments in VANETs based on traffic density. ACFM-TDMA leverages RSUs to schedule time slots adaptively, ensuring efficient channel utilization and fairness in medium access. While ACFM-TDMA improves packet delivery reliability, it does not incorporate energy-aware slot allocation, which is crucial for optimizing the power consumption of EVs in vehicular networks. To address this limitation, this research introduces a MAC protocol tailored for VANETs, where time is discretized into units termed "frames." Each frame is an aggregate of a predetermined quantity of "slots," facilitating structured temporal allocation for data transmission. The protocol operates on a cycle schedule maintained by road side nits (RSUs), with each cycle encompassing up to five frames ($N_F \leq 5$), aligning with the beaconing interval standard of VANET applications, typically 100 milliseconds (ms). The flow chart of the proposed ACFM-BMA method is shown in Figure 1. The flowchart shows that the operation starts with the RSU scheduling time slots, followed by vehicles contending for slots. Based on battery levels and buffer status, the RSU allocates slots through a priority-based mechanism. Vehicles with assigned slots proceed to transmit data, while others enter sleep mode to conserve energy. After transmission, the RSU collects feedback from vehicles and dynamically adjusts scheduling for the next cycle, ensuring adaptive and energy-efficient communication in VANETs.

Each frame is composed of one RSU slot ($T_{RSU} = T_c + G$), lasting 2 ms, followed by 36 data slots (T_d), each with a duration of 0.5 ms. The cycle's structure ensures that the cumulative period does not exceed 100 ms. Consequently, a full cycle can contain a maximum of five RSU slots and 180 data slots. In a dense traffic scenario, there will be 156 vehicles, a number lower than the available data slots (180), ensuring that the ACFM-TDMA protocol can allocate sufficient data slots to each vehicle, as outlined in [18]. Therefore, based on the earlier literature [18], we have considered 50 nodes as low, 100 nodes as medium, and 150 nodes as high vehicle traffic, as these thresholds align with commonly used benchmarks for evaluating network performance in vehicular environments.

The length of T_d within this protocol is a crucial system parameter that must be configured to ensure that each data slot (DS) is sufficiently long to transmit a complete beacon packet. The DS time is given by Equation (1):

$$T_d = \frac{(P_{Size} * 8)}{D_R} + G \tag{1}$$

Here the vehicular data packet size P_{Size} doesn't exceed 200 bytes, transmission rate D_R can reach up to 4 Mbps and the guard time G is still set to 100 μs given in [20]. So, T_d time allocated to a vehicle is 500 μs or 0.5 ms.

In adherence to U.S. IEEE standards [5], the protocol designates a guard time (G) of 100 μs between consecutive slots to mitigate inter-slot interference and sets a transmission rate (D_R) of 4 Mbps given in [21]. Therefore, the maximum packet size of an RSU control message (T_c) transmitted in an RSU segment can be given by Equation (2):

$$\begin{aligned} P_c &= \frac{1}{8}(T_{RSU} - G) * D_R \\ &= \frac{1}{8}(T_c - G) * D_R \end{aligned} \tag{2}$$

In the RSU segment, the time period $T_{RSU} = T_c + G = 2$ ms, with a transmission rate $D_R = 4$ Mbps, and a guard time $G = 100$ μs. The control packet in an RSU can span up to $P_c = 950$ Bytes, a sufficient capacity for transmitting the control message within a single frame, as discussed in [18].

In our proposed ACFM-BMA protocol, within this framework, two slots are specifically reserved for a contention period in each frame, as shown in Figure 2. Therefore, a full cycle will contain a maximum of 5 RSU slots, 10 contention slots, and 170 data slots.

In the contention period, vehicles intending to transmit convey their operational status using a two reserve bits, indicating:

- 00: Low battery and low buffer (2nd priority)
- 01: Low battery and high buffer (1st priority)
- 10: High battery and low buffer (4th priority)
- 11: High battery and high buffer (3rd priority)

The transmission of these packets signals to the controlling RSU that the vehicle possesses data awaiting transmission. The RSU then allocates a slot within the frame for the vehicle, ordering

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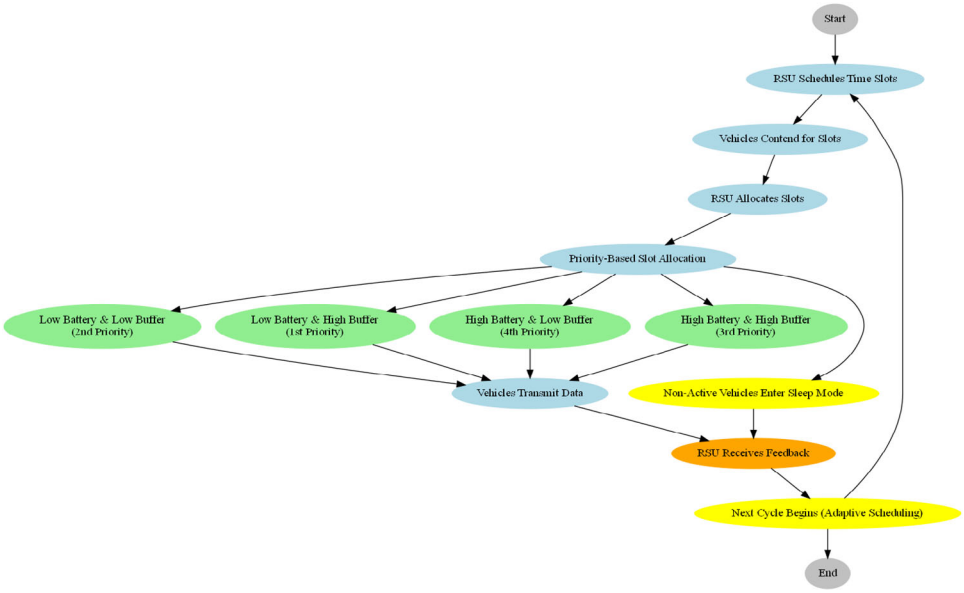

FIGURE 1 | Flow chart of the proposed ACFM-BMA MAC protocol.

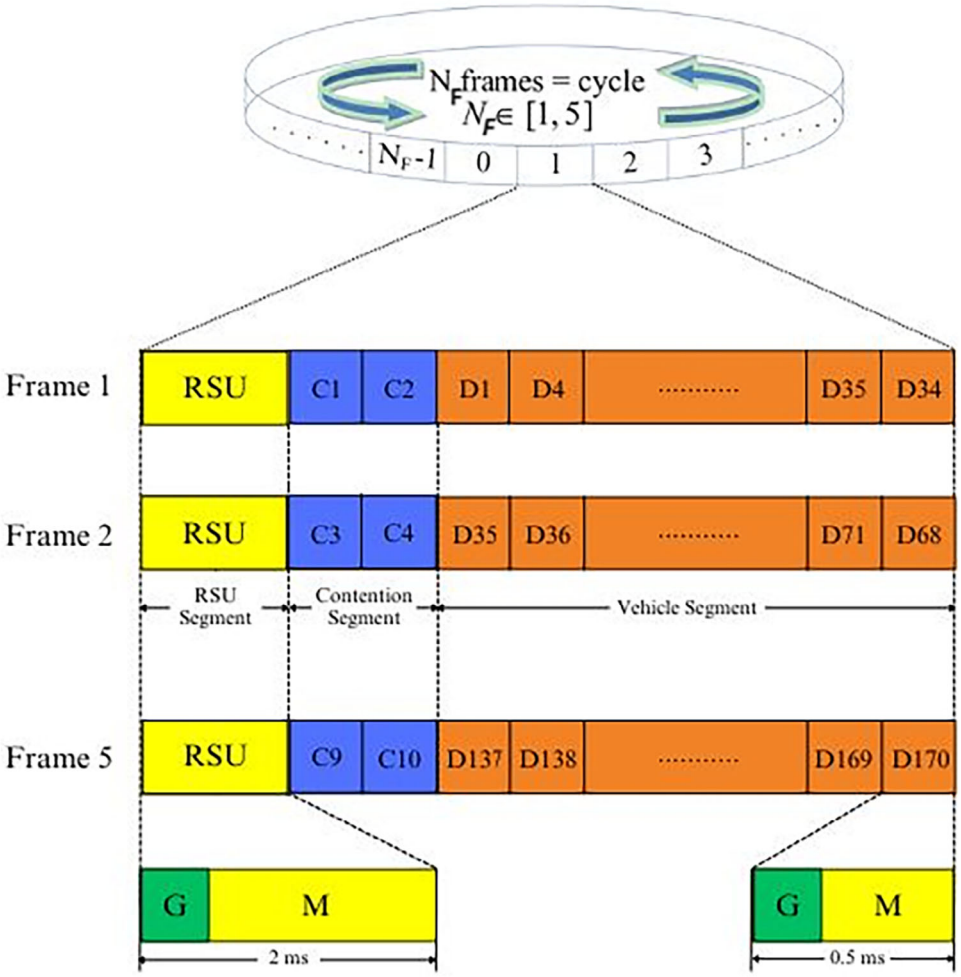

FIGURE 2 | Proposed contention based bit-mapping frame structure.

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the allocation based on the conveyed battery and buffer status. The inability to transmit the two-bit package indicates that the vehicle lacks data for transmission. Vehicles devoid of data for transmission conserve energy by deactivating their radios and reactivating them for the subsequent contention period.

Each RSU disseminates a control message at 20 ms intervals, consistently broadcasting a DS assignment schedule. This schedule is meticulously crafted to ensure that vehicles can transmit their beacons to adjacent vehicles without collision risks, thereby enhancing the reliability and efficiency of the communication network. For the performance analysis, the energy consumption model of both the existing ACFM-TDMA and the proposed ACFM-BMA MAC model is derived in the below subsection.

3.1 | Energy Consumption Model of ACFM-TDMA

The energy consumed at the RSU during the transmission of the control packet in an ACFM-TDMA system can be calculated using Equation (3):

$$EC_{ACFM-TDMA-CP-RSU} = P_t(T_c + G)N \tag{3}$$

Here, P_t represents the transmission power, T_c denotes the time taken to transmit the control packet (RSU segment), G stands for the guard band, N signifies the number of vehicles, and N_F is representing number of frames. During the contention period, all the vehicle nodes turn on their radio to receive the RSU control packet during the first frame. As already discussed in the case of TDMA, 36 vehicle nodes get data slots in each frame. Thus, in the next frame, only the remaining nodes turn on the radio during the control slot duration. The total number of vehicle nodes in each frame and total summation for all N_F frames are derived for total energy consumption estimation. The total energy consumed by the vehicles during the reception of the control packet in an ACFM-TDMA system can be calculated using Equation (4):

$$\begin{aligned} EC_{ACFM-TDMA-CP-V} &= P_r(T_c + G)\left(36\frac{N_F(N_F-1)}{2} + (N - 36(N_F-1))N_F\right) \\ &= P_r(T_c + G)(18N_F(N_F-1) + NN_F - 36N_F(N_F-1)) \\ &= P_r(T_c + G)(N_F(N - 18(N_F-1))) \end{aligned} \tag{4}$$

In this context, P_r represents the receiving power, and N_F can be calculated as $N_F = \lceil \frac{N}{36} \rceil$. The RSU unit receives data from vehicles during the data transmission period, continuous monitoring and event monitoring source vehicle nodes transmits data in their respective data slots, and other non-source nodes remain off during the data transmission duration. The energy consumed by the RSU during the reception of data from vehicles in the TDMA-based protocol can be calculated using Equation (5):

$$\begin{aligned} EC_{ACFM-TDMA-DS-RSU} &= nP_r(T_d + G) + mP_r(T_d + G) \\ &+ (N - m - n - 1)P_i(T_d + G) \end{aligned} \tag{5}$$

Here T_d represents the duration of the data slot (0.5 ms), and m and n correspond to the continuous traffic generating and event-oriented nodes, respectively. During the data transmission period, continuous monitoring and event monitoring source vehicle nodes turn on the radio in their respective data slots, and other non-source nodes remain off during the data transmission duration. The total energy consumed by vehicles during the data slot transmission in an ACFM-TDMA system can be expressed as Equation (6):

$$\begin{aligned} EC_{ACFM-TDMA-DS-V} &= (m + n)(P_t(T_d + G) + (N-2)P_i(T_d + G)) \\ &+ (N - m - n - 1)((N-1)P_i(T_d + G)) \end{aligned} \tag{6}$$

Here, P_i refers to the power consumed when idle.

3.2 | Energy Consumption Model of Proposed ACFM-BMA

The ACFM-BMA protocol is proposed as a superior alternative to the ACFM-TDMA protocol. In ACFM-BMA, the frame starts with the contention period. During the contention period, the RSU broadcasts a message to all the vehicle nodes, and after that, the contention period starts, during which vehicle devices transmit the bit-mapping status to the RSU as shown in Figure 2. The energy consumed at the RSU during the transmission of the control packet (RSU segment) in the BMA protocol is given by Equation (7):

$$EC_{ACFM-BMA-CP-RSU} = P_t(T_c + G)N \tag{7}$$

Here, P_t represents the transmission power, T_c denotes the time taken to transmit the control packet (RSU segment), G stands for the guard band, and N represents the number of vehicles. During the contention period, all the vehicle nodes turn on their radio to receive the RSU control packet during the first frame. As already discussed in the case of BMA, 34 vehicle nodes (2 data slots are reserved for the contention segment) get data slots in each frame. Thus, in the next frame, only the remaining nodes turn on the radio during the control slot duration. The total number of vehicle nodes in each frame and total summation for all N_F frames are derived for total energy consumption estimation. The total energy consumed by the vehicles during the reception of the control packet at the RSU for the proposed protocol is expressed as Equation (8):

$$\begin{aligned} EC_{ACFM-BMA-CP-V} &= P_r(T_c + G)\left(34\frac{N_F(N_F-1)}{2} + (N - 34(N_F-1))N_F\right) \\ &= P_r(T_c + G)(17N_F(N_F-1) + NN_F - 34N_F(N_F-1)) \\ &= P_r(T_c + G)(N_F(N - 17(N_F-1))) \end{aligned} \tag{8}$$

In this equation, P_r denotes the receiving power, and N_F is the calculated number of frames, which can be calculated as $N_F = \lceil \frac{N}{34} \rceil$. As shown in Figure 2, the contention segment is equal to $2 \times T_d$. The total energy consumed by the RSU during the contention

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TABLE 1 | Summary of parameters used in energy consumption calculations.

Notation	Parameter name	Value
P_t	Transmission power	0.891 W
P_r	Receiving power	0.0363 W
P_i	Idle mode power	0.0363 W
P_s	Sleep mode power	0.000363 W
T_c	Transmit the control packet (RSU segment) duration	1.9 ms (0.0019 s)
T_{co}	Bit-map status control packet duration	20 μs (0.00002 s)
T_d	Data slot duration	0.5 ms (0.0005 s)
G	Guard band	0.1 ms
N	Number of vehicles (Nodes)	15-150
N_F	Number of frames (RSU segments)	$\lceil \frac{N}{36} \rceil$ for TDMA, $\lceil \frac{N}{34} \rceil$ for BMA
m	Continuous traffic generating nodes	0.2 * N
p	Event generation probability	0-0.8
n	Event-oriented nodes	(N-m-1)*p

period for the proposed protocol is expressed as Equation (9):

$$EC_{ACFM-BMA-CAP-RSU} = P_r(2T_d + G)N_F \tag{9}$$

Here, T_d represents the duration of one data slot. The cumulative energy consumption of vehicles during the control packet transmission for the proposed protocol is stated as Equation (10):

$$EC_{ACFM-BMA-CAP-V} = NP_tT_{co} \tag{10}$$

Here T_{co} represents the duration of the bit-map control packet. The energy expended by the RSU during the reception of data from vehicles (nodes) within the BMA-based protocol can be determined utilizing Equation (11):

$$EC_{ACFM-BMA-DS-RSU} = nP_r(T_d + G) + mP_r(Td + G) + (N - m - n - 1)P_s(T_d + G) \tag{11}$$

where P_r denotes the receiving power, T_d represents the duration of the data slot (0.5 ms), G stands for the guard band, and m and n correspond to the continuous traffic generating and event-oriented nodes, respectively. The overall energy expended by vehicles during the transmission of data within a BMA system can be described as:

$$EC_{ACFM-BMA-DS-V} = (m + n)(P_i(T_d + G) + (N - 2)P_i(T_d + G)) \tag{12}$$

In this context, P_i signifies the power consumption during idle periods.

4 | Results Analysis and Discussion

To validate the performance of the proposed method, the energy consumption mathematical model is developed, and performance is analyzed in the MATLAB tool. For the energy consumption analysis, the SST11LF04 transceiver is considered [22]. As per the datasheet, the transmitter power (P_t) consumption is 0.270A @ 3.3 V (0.891 W), and the receiver power (P_r) consumption is 0.011 A @ 3.3 V (0.0363 W). The sleep mode power is very low as compared to the receiver power (approx 1% of P_r). The default parameters for the simulation analysis are given in Table 1. There are two different control packets are used for the control of overall operation. The first control packet is the RSU segment control packet ($T_c = 1.9$ ms), and the second control packet is the contention segment control packet ($T_{co} = 20\mu$s) used for bit-status.

The performance analysis is divided into three different scenarios. In the first scenario, the probability of event generation data traffic is varied. In the second scenario, the overall number of nodes is varied, and in the third scenario, periodic monitoring nodes are varied.

4.1 | Scenario-1: Event Occurrence Probability

In this scenario, the performance is analyzed for three different conditions. The network is analyzed for low (maximum 50 nodes), medium (maximum 100 nodes), and high (maximum 150 nodes) vehicle traffic. Studies have shown that networks with fewer than 50 nodes typically experience minimal congestion and low interference [18], whereas those with 100–150 nodes exhibit increasing levels of communication overhead, delay, and packet loss, making them suitable for assessing the scalability and efficiency of the proposed scheme under varying traffic densities. The results for low, medium, and high vehicle traffic conditions are shown in Figures 3, 4, and 5, respectively.

Figure 3 illustrates a comparative analysis of energy consumption between two medium access control (MAC) protocols, ACFM-TDMA and ACFM-BMA (proposed MAC), in scenarios with less vehicle traffic. The results indicate that the ACFM-BMA protocol consistently achieves lower energy consumption across

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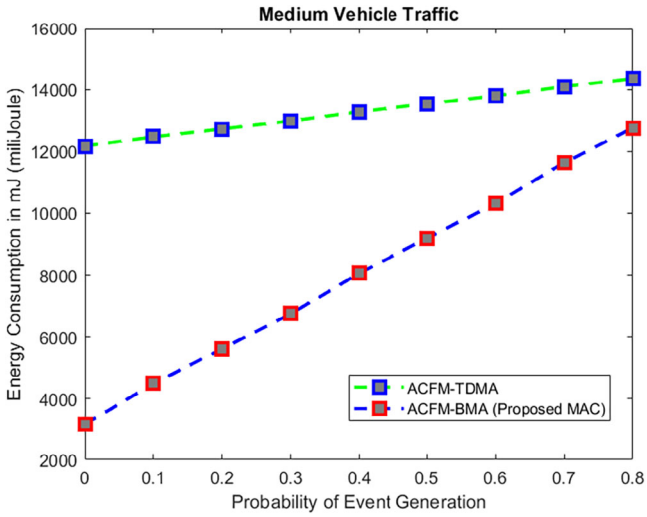

FIGURE 3 | Event probability variation for less vehicle traffic.

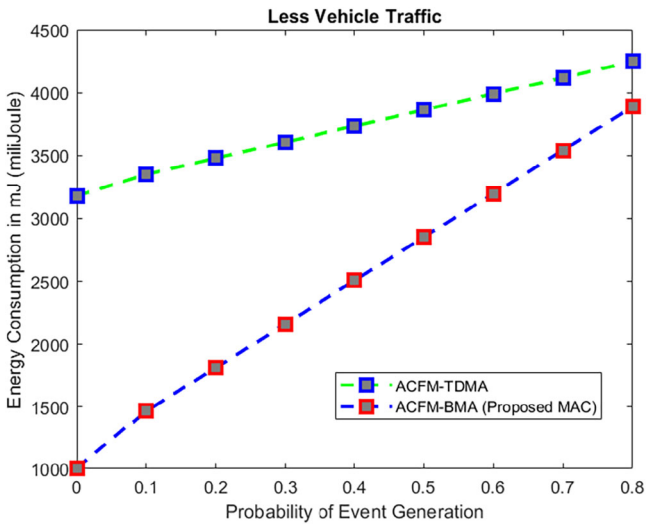

FIGURE 4 | Event probability variation for medium vehicle traffic.

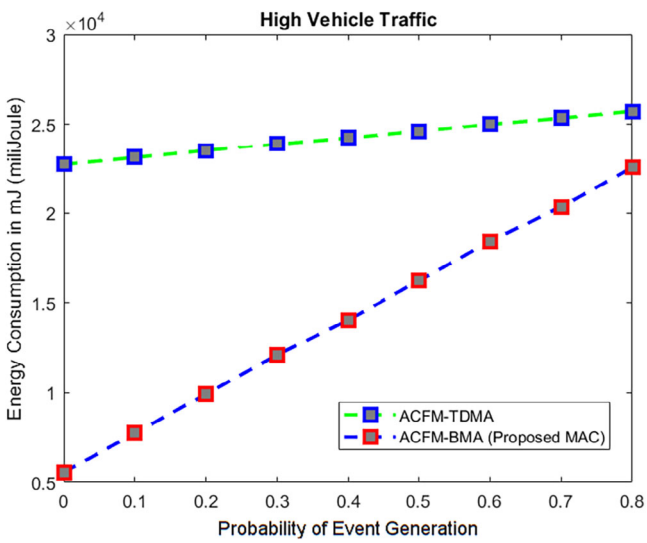

FIGURE 5 | Event probability variation for high vehicle traffic.

various probabilities of event generation compared to ACFM-TDMA. The energy saving is maximum when the event of data traffic generation probability is lowest. This is evident from the steeper slope and higher values associated with ACFM-TDMA, highlighting its less efficient energy utilization. The improved performance of ACFM-BMA is due to its strategic design, which focuses on energy conservation. In ACFM-BMA, only the source nodes actively reserve the channel using a contention segment, allowing non-source devices to remain in sleep mode throughout the communication process. Additionally, the roadside unit (RSU) also enters sleep mode during non-source node slots, further reducing unnecessary energy expenditure. This selective activation mechanism significantly cuts down on the overall energy consumption.

As shown in Figure 4, for medium vehicle traffic, ACFM-BMA shows lower energy consumption across varying probabilities of event generation, similar to the scenario with less vehicle traffic. The energy consumption for ACFM-TDMA increases more steeply, indicating less efficient energy utilization compared to ACFM-BMA. Figure 5 demonstrates that under high vehicle traffic conditions, the energy consumption for both ACFM-TDMA and ACFM-BMA protocols is substantially higher compared to medium and low traffic scenarios. While ACFM-BMA still maintains a lower energy consumption than ACFM-TDMA, the gap between the two protocols widens as the probability of event generation increases. This difference is more pronounced in high-traffic conditions, reflecting the greater efficiency of ACFM-BMA in conserving energy even as the communication load intensifies. This proves that the proposed method is highly scalable and suitable very high density traffic also.

The ACFM-BMA protocol integrates adaptive collision-free techniques with a bit map-assisted approach, taking into account the electric vehicle (EV) battery levels and buffer status. This allows for optimized communication efficiency by dynamically managing the communication channel based on real-time data about battery levels and buffer space availability. As a result, the protocol ensures effective data transmission while maximizing resource utilization and conserving energy. The proposed approach leads to a balanced performance, enhancing the overall efficiency of vehicular communication systems, and proves a clear advantage over traditional methods like ACFM-TDMA.

4.2 | Scenario-2: Vehicle Traffic

In this scenario, again, three conditions are analyzed, that is, low ($p = 0.2$), medium ($p = 0.5$), and high ($p = 0.8$) event data traffic generation probability. The results are shown in Figures 6–8.

Figure 6 represents a comparative analysis of energy consumption between ACFM-TDMA and ACFM-BMA (proposed MAC) with less event generation data traffic. The results indicate that the ACFM-BMA protocol consumes less energy for low to high-density vehicle traffic compared to ACFM-TDMA. The traffic is varied from 15 to 150 vehicles. The energy saving is maximum when the number of vehicles is maximum. This is due to the fact that the event probability is less in the first case, and hence, the source nodes' ratio decreases as compared to the non-source nodes.

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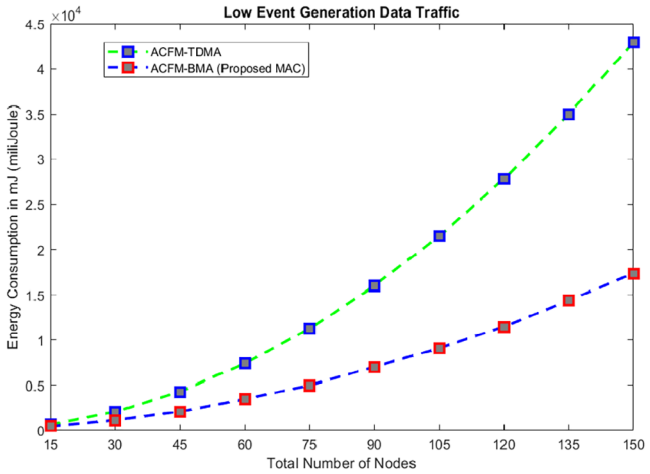

FIGURE 6 | Total no. of vehicles variation for low event generation data traffic.

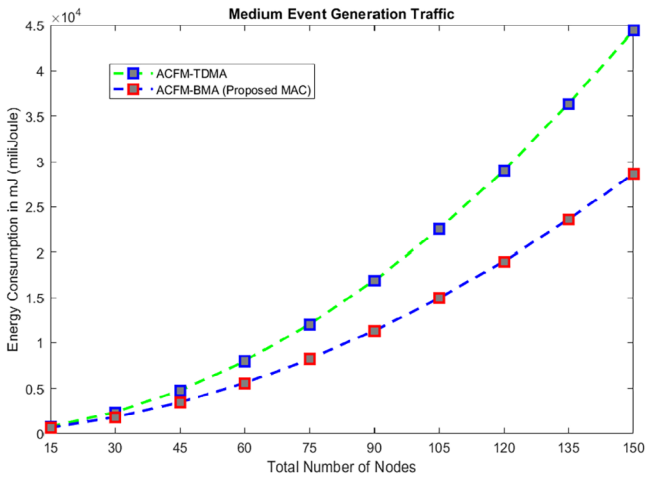

FIGURE 7 | Total no. of vehicles variation for medium event generation data traffic.

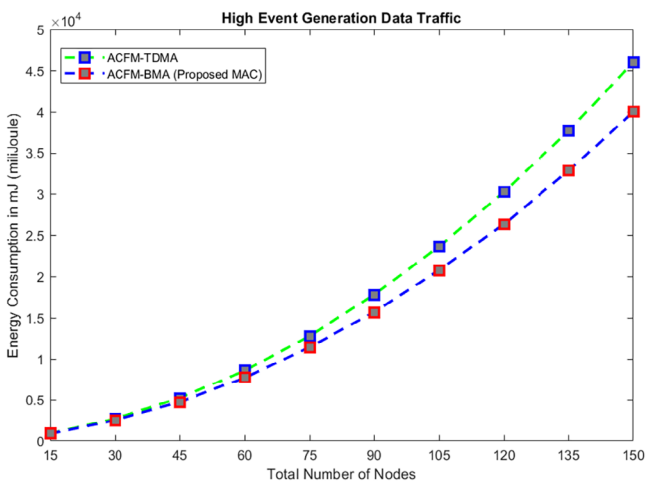

FIGURE 8 | Total no. of vehicles variation for high event generation data traffic.

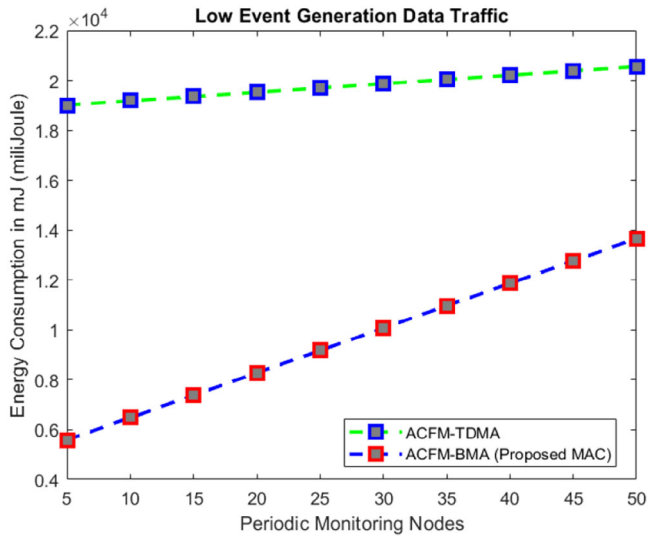

FIGURE 9 | Periodic monitoring vehicles variation for low event generation data traffic.

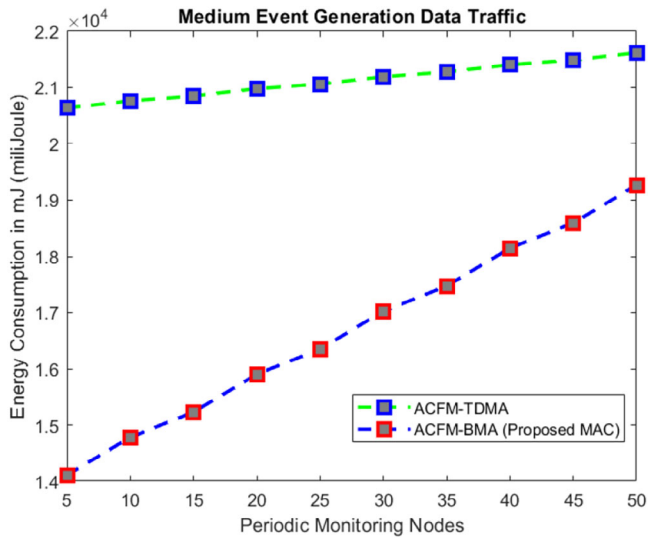

FIGURE 10 | Periodic monitoring vehicles variation for medium event generation data traffic.

Figures 7 and 8 represent a comparative analysis for medium and high event generation data traffic. The results indicate that maximum energy savings are lower with medium event probability and even less with high event probability. This trend occurs because, as event probability increases, the proportion of source nodes rises relative to non-source nodes.

4.3 | Scenario-3: Periodic Monitoring Nodes

In this scenario also, three conditions are analyzed, that is, low ($p = 0.2$), medium ($p = 0.5$), and high ($p = 0.8$) event data traffic generation probability. The results are shown in Figures 9–11.

In this scenario, the number of nodes that generate data traffic periodically varied from 5 to 50 and the performance is compared for various event probabilities in three different plots, that is,

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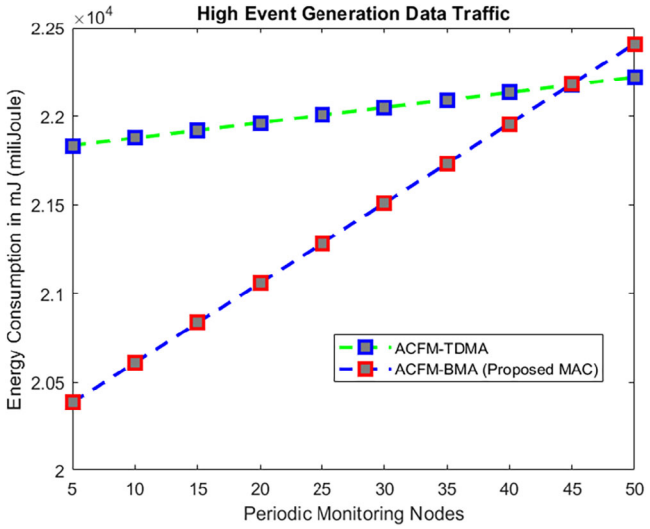

FIGURE 11 | Periodic monitoring vehicles variation for high event generation data traffic.

Figures 9–11. In the first case, energy consumption is analyzed for the low event probability, as shown in Figure 9. The energy consumption for both protocols increases as the number of monitoring nodes grows. However, the ACFM-BMA protocol consistently exhibits lower energy consumption than the ACFM-TDMA protocol, with the difference becoming more significant as the number of nodes increases. This indicates that ACFM-BMA is more energy-efficient, particularly in scenarios with a higher number of monitoring nodes. Similar to the low event generation scenario, in the medium event generation case, as shown in Figure 10, both protocols show increased energy consumption with more monitoring nodes. However, the ACFM-BMA protocol consistently demonstrates lower energy consumption than ACFM-TDMA, though the difference between the two is less pronounced compared to the low event generation case. This indicates that while ACFM-BMA remains more efficient, its energy-saving advantage diminishes as the event generation probability increases. In the last case as shown in Figure 11, the energy consumption for both protocols increases as the number of monitoring nodes rises. However, unlike in low and medium event generation scenarios, ACFM-BMA exhibits higher energy consumption than ACFM-TDMA in this high event generation context. The difference in energy consumption between the two protocols becomes more significant as the number of monitoring nodes increases. This indicates that while ACFM-BMA was more efficient in lower event generation scenarios, its energy-saving advantage diminishes and reverses in high event generation conditions, making ACFM-TDMA the more energy-efficient protocol under these circumstances.

5 | Conclusion and Future Work

In the present work, the energy consumption of the proposed ACFM-BMA MAC protocol is analyzed and compared with the existing ACFM-TDMA MAC protocol. The performance is analyzed into three different scenarios. In the first scenario, For high vehicle traffic, ACFM-BMA shows a substantial energy saving, achieving up to 45% reduction in energy consumption compared to ACFM-TDMA. In medium vehicle traffic scenarios, the energy saving is slightly lower but still notable, with up to 40% reduction. For low vehicle traffic, the energy saving is approximately 35%. These results demonstrate that the proposed ACFM-BMA protocol consistently outperforms ACFM-TDMA, providing significant energy savings across various traffic conditions. In the second scenario, the most significant energy savings occur under high vehicle traffic and lower event generation probabilities. As event probability increases, energy savings decrease due to the higher proportion of source nodes. ACFM-BMA's performance advantage is evident across varying traffic densities and event probabilities. This makes ACFM-BMA particularly suitable for environments with low to medium event generation. In the last scenario, in low event generation data traffic, ACFM-BMA exhibits significant energy savings of approximately 30%. In medium event generation data traffic (5–50 nodes), ACFM-BMA maintains a smaller but still notable energy saving of around 6.8%. However, in high event generation conditions, ACFM-BMA becomes less efficient than ACFM-TDMA, with a slight increase in energy consumption by 2.3% at 50 nodes. The improved performance of ACFM-BMA is due to its selective activation mechanism, where only source nodes and RSUs engage during communication, allowing non-source nodes to remain in sleep mode. This strategy significantly reduces overall energy consumption by minimizing unnecessary activity. In the future, the proposed MAC protocol can be improved by efficiently reducing overhead control packets.

Author Contributions

M.T. and K.T.V. contributed to the writing, review, and editing of the manuscript. P.K. was responsible for proofreading and editing of final manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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